lec 4

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Chapter 3 Outline

- The Relational Data Model and Relational Database Constraints
- Relational Model Constraints and Relational Database Schemas
- Update Operations, Transactions, and Dealing with Constraint Violations

The Relational Data Model and Relational Database Constraints

- Relational model
  - First commercial implementations available in early 1980s
  - Has been implemented in a large number of commercial system
- Hierarchical and network models
  - Preceded the relational model(This means that the hierarchical and network database models were developed and used before the relational model became widely adopted.)

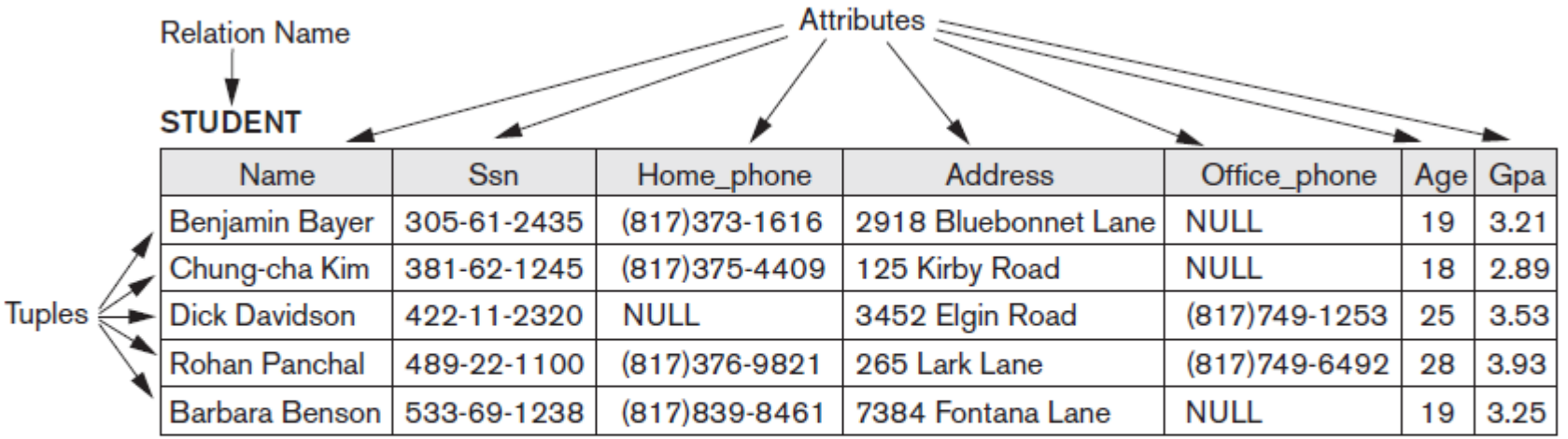
Relational Model, Hierarchical Model and Hierarchical Model

- Relational Model: Stores data in tables consisting of rows and columns. Tables are linked using keys. Example: MySQL.
- - Hierarchical Model: Organizes data in a tree structure, where each child record has only one parent. Example: A university → Faculties → Departments.
- - Network Model: Organizes data as a network, where a child record can have multiple parent records. Example: A student can enroll in multiple courses, and each course can have multiple students.
- Key difference: The relational model uses tables, the hierarchical model uses a tree structure, and the network model uses interconnected records.

Relational Model Concepts

- Represents data as a collection of relations
- **Table** of values
  - Row
    - Represents a collection of related data values
    - Fact that typically corresponds to a real-world entity or relationship
    - *Tuple is a row in a relational database table.*
  - Table name and column names
    - Interpret the meaning of the values in each row *attribute*

Relational Model Concepts (cont'd.)



Name	Ssn	Home_phone	Address	Office_phone	Age	Gpa
Benjamin Bayer	305-61-2435	(817)373-1616	2918 Bluebonnet Lane	NULL	19	3.21
Chung-cha Kim	381-62-1245	(817)375-4409	125 Kirby Road	NULL	18	2.89
Dick Davidson	422-11-2320	NULL	3452 Elgin Road	(817)749-1253	25	3.53
Rohan Panchal	489-22-1100	(817)376-9821	265 Lark Lane	(817)749-6492	28	3.93
Barbara Benson	533-69-1238	(817)839-8461	7384 Fontana Lane	NULL	19	3.25

Figure 3.1
The attributes and tuples of a relation STUDENT.

Domains, Attributes, Tuples, and Relations

- **Domain** D
  - Set of atomic values
  - Example: The domain of Grade may be numbers from 0 to 100.
- **Atomic**
  - Each value indivisible, meaning it is treated as a single value.
  - Example: Age = 20 is atomic
- Specifying a domain
  - **Data type** specified for each domain
  - such as INTEGER, VARCHAR, or DATE.

Domains, Attributes, Tuples, and Relations (cont'd.)

- **Relation schema** R
  - Denoted by $R(A_1, A_2, ..., A_n)$
  - Made up of a relation name R and a list of attributes, $A_1, A_2, ..., A_n$
- Attribute A_i
  - Name of a role played by some domain D in the relation schema R
- **Degree** (or **arity**) of a relation
  - Number of attributes n of its relation schema

Domains, Attributes, Tuples, and Relations (cont’d.)

- Relation (or Relation State): A set of rows (tuples) in a database table at a particular time.
- - Set of n-tuples: A relation contains multiple rows, and each row has (n) values, one for each column.
- - Each n-tuple: An ordered list of values representing one record.
- - Each value (v_i): Each value must belong to the domain of its corresponding attribute, or it can be NULL if the value is unknown or missing.
- A relation is a set of tuples (rows), and each tuple contains an ordered value for every attribute (column).

Domains, Attributes, Tuples, and Relations (cont’d.)

- Relation (r(R)): A table containing rows (tuples) at a particular time.
- Mathematical relation of degree (n): A relation has (n) attributes (columns), each with its own domain of valid values.
- Cartesian product ((X)): The set of all possible combinations of values from the specified domains.
- Subset ((⊆)): The actual relation contains only some of these possible combinations.
- dom(Name) = {Ahmed, Mona}
- dom(Age) = {20, 25}
- Their Cartesian product contains four possible combinations:
- & (Ahmed,20), (Ahmed,25)
- (Mona,20), (Mona,25)

Domains, Attributes, Tuples, and Relations (cont'd.)

- **Cardinality**
  - Total number of values in domain
  - Suppose the domain of the Gender attribute is:
  - D={{Male},{Female}}
  - The cardinality of this domain is 2, because it contains two possible values.
- **Current relation state**
  - Relation state at a given time
  - Reflects only the valid tuples that represent a particular state of the real world
- Attribute names
  - Indicate different **roles**, or interpretations, for the domain

Characteristics of Relations

- Ordering of tuples in a relation
  - Relation defined as a set of tuples
  - Elements have no order among them
- Ordering of values within a tuple and an alternative definition of a relation
  - Order of attributes and values is not that important
  - As long as correspondence between attributes and values maintained

Characteristics of Relations (cont'd.)

- Alternative definition of a relation
  - Tuple considered as a set of (<attribute>, <value>) pairs
  - Each pair gives the value of the mapping from an attribute A_i to a value v_i from dom(A_i)
- Use the first definition of relation
  - Attributes and the values within tuples are ordered
  - Simpler notation

Characteristics of Relations (cont'd.)

Figure 3.2
The relation STUDENT from Figure 3.1 with a different order of tuples.

STUDENT

Name	Ssn	Home_phone	Address	Office_phone	Age	Gpa
Dick Davidson	422-11-2320	NULL	3452 Elgin Road	(817)749-1253	25	3.53
Barbara Benson	533-69-1238	(817)839-8461	7384 Fontana Lane	NULL	19	3.25
Rohan Panchal	489-22-1100	(817)376-9821	265 Lark Lane	(817)749-6492	28	3.93
Chung-cha Kim	381-62-1245	(817)375-4409	125 Kirby Road	NULL	18	2.89
Benjamin Bayer	305-61-2435	(817)373-1616	2918 Bluebonnet Lane	NULL	19	3.21

Characteristics of Relations (cont'd.)

- Values and NULLs in tuples
  - Each value in a tuple is atomic
  - **Flat relational model**
    - Composite and multivalued attributes not allowed
    - **First normal form** assumption
  - Multivalued attributes
    - Must be represented by separate relations
  - Composite attributes
    - Represented only by simple component attributes in basic relational model

Characteristics of Relations (cont'd.)

- NULL values
  - Represent the values of attributes that may be unknown or may not apply to a tuple
  - Meanings for NULL values
    - *Value unknown*
    - *Value exists but is not available*
    - *Attribute does not apply to this tuple (also known as value undefined)*

Characteristics of Relations (cont’d.)

- Interpretation (meaning) of a relation
  - **Assertion**
    - Each tuple in the relation is a **fact** or a particular instance of the assertion
  - **Predicate**
    - Values in each tuple interpreted as values that satisfy predicate

Relational Model Notation

- Relation schema R of degree n
  - Denoted by $R(A_1, A_2, \ldots, A_n)$
- Uppercase letters Q, R, S
  - Denote relation names
- Lowercase letters q, r, s
  - Denote relation states
- Letters t, u, v
  - Denote tuples

Relational Model Notation

- Name of a relation schema: STUDENT
  - Indicates the current set of tuples in that relation
- Notation: STUDENT(Name, Ssn, ...)
  - Refers only to relation schema
- Attribute *A* can be qualified with the relation name *R* to which it belongs
  - Using the dot notation *R.A*

Relational Model Notation

- *n-tuple t* in a relation *r*(*R*)
  - Denoted by $t = \langle v_1, v_2, \ldots, v_n \rangle$
  - v_i is the value corresponding to attribute A_i
- Component values of tuples:
  - $t[A_i]$ and $t.A_i$ refer to the value v_i in t for attribute A_i
  - $t[A_u, A_w, \ldots, A_z]$ and $t.(A_u, A_w, \ldots, A_z)$ refer to the subtuple of values $\langle v_u, v_w, \ldots, v_z \rangle$ from t corresponding to the attributes specified in the list

Relational Model Constraints

- Constraints
  - Restrictions on the actual values in a database state
  - Derived from the rules in the miniworld that the database represents
- **Inherent model-based constraints** or **implicit constraints**
  - Inherent in the data model
  - Inherent model-based constraints (Implicit Constraints) are rules that are automatically part of a data model. They define what is allowed based on the structure and rules of that model.
  - Example in the relational model:
  - - A table contains rows and columns.
  - - Each cell contains a single atomic value.
  - - Duplicate rows are not allowed in a mathematical relation.
  - Remember: These constraints are built into the data model itself, rather than being defined separately by the database designer.